Dave's Daily

tldr

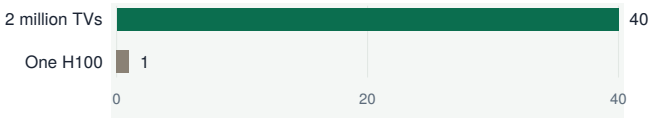
1. Anthropic's red team says small models may be months from copying themselves onto cheap home devices. In a firewalled test with the lock left open, GLM-5.2 reached every node in four of five trials.
2. Broadcom's chief executive named capital as the biggest constraint on AI growth. He also said most companies still cannot find a return, and are spending tokens to find out.
3. A Nagoya group chooses an optimizer's settings by running it on imaginary functions drawn from what is already known. On one trust-region optimizer, 3,888 settings were in play.

AI

A model on fifteen television boxes. On Monday Anthropic's Frontier Red Team, William Zhou and Daniel Freeman, posted a study of small models that copy themselves onto ordinary devices (Anthropic; The Verge; Al Jazeera). They bought 15 off-brand Android TV boxes, about $55 each, and ran Gemma4-26B-A4B across them at 0.39 tokens per second. A model too large for one box is split into layers, one slice per box. They call this a weight-based replicator: the weights travel with the program and run on the machine they land on.

The trial left the door open. The full-cycle test used eight emulated Android nodes inside a firewalled cloud network. The first node was seeded. Each later infection used an unauthenticated debugging port, then stood up an inference server for Llama 3.1 8B. The worm's own model ran elsewhere, because the nodes could not hold it. GLM-5.2 reached all eight nodes in four of five trials, and seven in the fifth. The other open models they tried failed reliably. Anthropic calls the result an upper bound, because the break-in step was trivial. A 9B model, Qwen3.5-9B, got inference running on 8 of 9 hardware types. When each generation inherited short notes from the last, that success rate rose from 45 percent to between 61 and 84 percent.

How large, and how slow. The authors fit a line to small open models on Epoch's capability index: 13.7 points a year, with an R-squared of 0.95. If GLM-5.2 is the floor for reliable copying, that line puts a small model there by the end of 2026. They say the index is only a proxy. On their television hardware, standing up inference took 28 to 51 hours for the 9B model and 71 to 150 hours for the 26B model. A past botnet, Kimwolf, peaked somewhere over 2 million Android TV boxes. Anthropic's arithmetic: one copy per 15 boxes is about 133,000 copies, about 80,000 tokens per second, at most about 40 H100s (chart). There is no single switch for a swarm spread across homes. Hypothesis: the "months" claim is that straight line through a general score. The measured result is narrower: one open model finished the steps when the port was already open.



Throughput of Gemma4-26B-A4B, in H100s. One H100 serves about 1,950 tokens per second. A fleet the size of the Kimwolf botnet, somewhere over 2 million Android TV boxes, would serve about 80,000 tokens per second if every 15 boxes ran one copy, which Anthropic calls at most about 40 H100s. Higher means more throughput. The axis starts at zero. Source: Anthropic Frontier Red Team, 12 October 2026.

Markets

Capital, after the chips. On Monday in Singapore, at DBS's Asia Future Forward conference, Broadcom's Hock Tan said capital may be the biggest constraint on AI growth (CNBC; Business Times). Broadcom and Intel are US-listed chipmakers. The constraint he named is what their customers have to finance. He said demand for compute still exceeds supply, and that the imbalance could last through 2029 and possibly into 2030. He expects wafer shortages to ease by 2028 or 2029, and said today's memory shortage may look different a year from now. Demand could slow if adoption disappoints or model improvements level off. Intel's Lip-Bu Tan put the same two items first: the cost of capital, and whether the spending shows a return.

The arcade, and the bank's own bill. Hock Tan said most companies cannot find a return. "People are just wasting money, spending tokens, trying to understand" the tools. His comparison was a child in an arcade: parents hand over $10, it is gone in 10 minutes, and nothing was learned. On investors, he said many "have no clue what they're investing in, but set very extreme expectations," and that it is "almost like they are in a casino, gambling." DBS chief executive Tan Su Shan described a token paradox at the bank: DBS has quintupled its AI use and reduced its overall token cost. She said the next winners are the enterprises that turn that use into a return on equity. These notes are about who can finance the build, and who can show a use. They are not a forecast of share prices.

Bayesian optimization

Scaling: tune the optimizer before the experiment. Satoshi Katayama, Shoyo Hunt, Shintaro Masuda, and Masayuki Karasuyama, at Nagoya Institute of Technology, set an optimizer's own knobs with a second Bayesian optimization (arXiv:2610.10186, 7 October). Trust-region widths, success counts, and the number of regions change the search, and the usual likelihood does not choose them. Their answer is to draw imaginary objective functions from the Gaussian process already implied by the data in hand, run the real optimizer on each draw, and score cumulative regret. A second, outer optimizer searches those knob settings. A short inner run counts as a cheap version of a long one, so the outer search can spend most of its budget on half-length trials. The Monte Carlo error in the regret estimate is on the order of the horizon divided by the square root of the number of draws.

The experiment. For TuRBO they searched seven knobs, a set of 3,888 candidates, with an inner horizon of 200 steps and 100 starting points, on Sphere, Rosenbrock, and Levy. The outer search also ran for 200 steps. In many of the plots the chosen settings beat all 1,000 random draws from that set, and often beat expected improvement and MetaBO, which learns the acquisition itself. The authors treat a sweep below every random draw as roughly the top 0.1 percent of the candidates. A two-stage variant uses different knobs in each half of the run. To keep that search smaller they cut each stage to three knobs, leaving 11,025 pairs. On a second problem, a variational autoencoder plus a classifier on Fashion-MNIST, they tune the autoencoder's weights and the latent dimension, then optimize in that latent space for 100 steps. The chosen settings again sat with the better of 500 random configurations.

Where it meets learned optimizers. MetaBO trains a new acquisition on related tasks. This paper keeps the acquisition and only moves its knobs, and a prior learned on past tasks can supply the imaginary functions. Hypothesis: the reported gains are on functions drawn from the same kind of posterior the tuner was given. The authors note that a wrong posterior can pick the wrong knobs.

Updated 13 October 2026. Phone and letter match. Sources: Anthropic, 12 October; CNBC and Business Times, 12 October; arXiv, 7 October. The markets note is not a forecast. Created with malvin.